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Breather tube for labyrinth seal chamber

Abstract

An electrical submersible pumping assembly includes a motor, a pump connected to the motor by a shaft, and a labyrinth type seal section between the motor and pump. Hydrostatic pressure is communicated between the motor and pump across the seal section and through a breather tube in the seal section that is configured to restrict fluid communication from the pump to the motor. A U-shaped bend is provided in the breather tube to increase its effective length. Optionally, a number of recycle loops are formed on the breather tube similar to a Tesla valve. In an alternative, the shaft includes a helical flight that directs fluid to a mechanical shaft seal between pump and seal section.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	**Issued Date**	**Patentee Name**	**U.S. Cl.**	**CPC**
1329559	12/1919	Tesla	137/842	F15D 1/02
3182214	12/1964	Boyd	277/408	F04D 29/128
4421999	12/1982	Beavers	310/87	F04D 13/10
4667737	12/1986	Shaw	166/104	F16J 15/004
4992689	12/1990	Bookout	N/A	N/A
5203080	12/1992	Bookout	29/890.09	H02K 5/132
5367214	12/1993	Turner, Jr.	N/A	N/A
6307290	12/2000	Scarsdale	310/87	H02K 5/132
6412562	12/2001	Shaw	166/105	F04D 13/086
6602059	12/2002	Howell	417/424.2	F04B 47/06
6666664	12/2002	Gross	417/424.2	F04D 13/10
7066248	12/2005	Howell	166/66.4	F04D 13/12
7367400	12/2007	Howell	166/66.4	H02K 5/132
8419387	12/2012	Karbs	417/414	F04B 53/02
9249649	12/2015	Fripp	N/A	E21B 34/08
9303648	12/2015	Mack	N/A	N/A
9695654	12/2016	Stephenson	N/A	E21B 21/01
10299636	12/2018	Ophardt	N/A	B01F 25/432
10302089	12/2018	Meyer	N/A	F04D 13/08
10371167	12/2018	Meyer	N/A	F04D 13/10
10519756	12/2018	Crane	N/A	E21B 41/0085
10822933	12/2019	Crane	N/A	F04D 29/586
10928841	12/2020	Tanner	N/A	E21B 43/128
11365809	12/2021	Collins	N/A	F04D 29/106
11493048	12/2021	MacIver	N/A	F04D 13/086
12196209	12/2024	Koester	N/A	F04C 23/008
2007/0207046	12/2006	Du	310/87	E21B 43/128
2011/0236233	12/2010	Merill	417/410.1	F04B 17/03
2014/0069629	12/2013	McCann	166/66.4	E21B 43/128
2016/0076550	12/2015	Collins	277/336	F16J 15/002
2017/0037861	12/2016	Rutter	N/A	F04D 29/102
2018/0313359	12/2017	Meyer	N/A	E21B 43/128
2019/0003476	12/2018	Jee	N/A	F04B 53/18
2020/0072357	12/2019	MacIver	N/A	E21B 43/128
2023/0174870	12/2022	Presley	585/241	C10G 1/10
2024/0102355	12/2023	Dorban	N/A	E21B 43/12
2024/0337169	12/2023	Romer	N/A	E21B 33/1208

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Background/Summary

BACKGROUND OF THE INVENTION

1. Field of Invention

(1) The present disclosure relates to an electrical submersible pump system ("ESP"), and more particularly to an ESP having a labyrinth seal section with a breather tube that substantially limits flow to a single direction.

2. Description of Prior Art

(2) Artificial lift is generally employed in hydrocarbon producing wells that lack adequate pressure to lift liquid from inside the well. One type of artificial lift is an electrical submersible pump ("ESP") that includes an electrically powered motor filled with a dielectric fluid. A drive shaft connects the motor to a pump, energizing the motor rotates the shaft that rotates impellers in the pump. The pump is often a centrifugal pump with multiple stages of impellers and diffusers for pressurizing the liquid. Typically, a seal section is included between the motor and the pump for equalizing pressure of the dielectric fluid inside the motor with hydrostatic pressure in the well. If well fluids are allowed to enter the motor an electrical short can occur and damage the motor.

(3) Seal sections are typically designed to communicate hydrostatic pressure to dielectric fluid in the motor without allowing wellbore fluid to enter the motor. The seal sections usually include either a flexible diaphragm, a bellows, or a labyrinth chamber. In diaphragms and bellows the physical barriers between the wellbore fluid and dielectric fluid are elastomer or metal membranes, and in labyrinth chambers the physical barrier is an elongated flow circuit filled with dielectric fluid Wellbore fluid can still migrate to the motor if the labyrinth chamber is an inadequate obstacle, or failure of seals on the drive shaft within the seal section.

SUMMARY OF THE INVENTION

(4) Disclosed herein is an example of an electrical submersible pumping system ("ESP") that includes a motor, a shaft having an end attached to the motor, a pump attached to an end of the shaft distal from the motor, and a seal section disposed between the motor and the pump. In this example the seal section includes a breather tube having, an uphole end that is in pressure communication to ambient hydrostatic pressure, a downhole end distal from the uphole end and that is in pressure communication with the motor, and a mid-portion having an undulating contour. Examples of the undulating contour include a U-shaped bend. In an alternative, the undulating contour is a recycle loop mounted to the breather tube, the recycle loop having a curved mid-section and upper and lower ends that intersect the breather tube at spaced apart locations along a length of the breather tube to define a valvular conduit. Optionally, the upper end intersects the breather tube at an upper intersection that is uphole of where the lower end intersects the breather tube, where the portion of the recycle loop proximate to the upper intersection is at an acute angle to a portion of the breather tube adjacent the upper intersection and on a side opposite the uphole end, and where the portion of the recycle loop proximate to the lower intersection is at an acute angle to a portion of the breather tube adjacent the lower intersection and on a side opposite the uphole end. In another example, the ESP further includes a multiplicity of recycle loops arranged in series along the length of the breather tube. The ESP optionally includes a seal section housing, a guide tube in the seal section that circumscribes a portion of the shaft and that is spaced radially inward from the seal section housing, and a labyrinth chamber defined in annulus between the guide tube and seal section housing. In this example the breather tube is disposed in the labyrinth

chamber and a port is optionally formed radially through the guide tube at a location axially between the uphole and downhole ends of the breather tube, so that a pressure communication path between the pump and the motor extends from the uphole end of the breather tube to the downhole end of the breather tube, uphole to the port, and downhole to the motor in an annular space between the shaft and the guide tube. In another example the ESP includes a mechanical seal on the shaft and a helical flight on the downhole of the mechanical seal, the flight configured to increase fluid pressure on a downhole side of the mechanical seal. Alternatives of undulating contour include a diameter of the breather tube that increases with distance from uphole end to the downhole end.
(5) Another example of an electrical submersible pumping system ("ESP") is disclosed and that includes a motor, a pump uphole of the motor, a shaft having an end attached the motor and an opposing end attached to the pump, and a seal section disposed between the motor and the pump having a labyrinth chamber with a flow path that restricts fluid communication. The ESP of this example further optionally includes a helical flight on an outer surface of the shaft that operates as a positive displacement pump with rotation of the shaft and so that fluid exiting the positive displacement pump generates an increase in pressure on a downhole side of a mechanical seal around the shaft. In an example the seal section includes an outer housing with a seal head on an uphole end of the housing, the seal head having an axial bore that receives the shaft, and wherein a vent tube is formed through the seal head that provides a flow path for fluid to flow into the seal section from adjacent the mechanical seal. In alternatives the flow path extends through a breather tube with a U-shaped portion, a breather tube with a diameter that increases with distance downhole, or a valvular conduit. In one example, fluid flow in a direction to the motor is restricted in the valvular conduit, and wherein fluid flow in a direction away from the motor flows substantially unimpeded through the valvular conduit.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) Some of the features and benefits of the present invention having been stated, others will become apparent as the description proceeds when taken in conjunction with the accompanying drawings, in which:
(2) FIG. **1** is a side sectional view of an example of an Electrical Submersible Pumping Assembly ("ESP") disposed in a wellbore.
(3) FIG. **2** is a side sectional view of an example of a portion of a seal section in the ESP of FIG. **1**.
(4) FIGS. **2**A-**2**C are side schematic views of an embodiment of a breather tube in the seal section of FIG. **2**.
(5) FIG. **3** is a side sectional view of an alternate example of the seal section of FIG. **2**.
(6) FIG. **3**A is an axial view of a portion of FIG. **3** taken along lines **3**A-**3**A.
(7) While subject matter is described in connection with embodiments disclosed herein, it will be understood that the scope of the present disclosure is not limited to any particular embodiment. On the contrary, it is intended to cover all alternatives, modifications, and equivalents thereof.
DETAILED DESCRIPTION OF INVENTION
(8) The method and system of the present disclosure will now be described more fully hereinafter with reference to the accompanying drawings in which embodiments are shown. The method and system of the present disclosure may be in many different forms and should not be construed as limited to the illustrated embodiments set forth herein; rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey its scope to those skilled in the art. Like numbers refer to like elements throughout. In an embodiment, usage of the term "about" includes +/−5% of a cited magnitude. In an embodiment, the term "substantially" includes +/−5% of a cited magnitude, comparison, or description. In an embodiment, usage of the term

“generally” includes +/−10% of a cited magnitude.
(9) It is to be further understood that the scope of the present disclosure is not limited to the exact details of construction, operation, exact materials, or embodiments shown and described, as modifications and equivalents will be apparent to one skilled in the art. In the drawings and specification, there have been disclosed illustrative embodiments and, although specific terms are employed, they are used in a generic and descriptive sense only and not for the purpose of limitation.
(10) Show in a side partial sectional view in FIG. **1** is an example of an electrical submersible pumping system (“ESP”) **10** disposed in a well or **12** and which is used to artificially lift fluid F produced from surrounding formation **14**. In the example shown, perforations **16** are formed radially outward into formation **14** to allow a flow path for fluid F to make its way to wellbore **12**. ESP **10** includes a motor **18** on its lower or downhole portion. A pump **20** is on an upper end of ESP **10** located uphole from motor **18**. A seal section **22** is disposed between the motor **18** and pump **20**. Shown in dashed outline is a shaft **24** that extends from motor **18** to pump **20** and axially through the seal section **22**. Fluid F includes liquid L, which is shown having accumulated to a level within wellbore **12**. The level of the liquid L is above an inlet **26** to the pump **20**, and when pump **20** is operating the fluid F is drawn into pump **20** via inlet **26**. Inside pump **20** are impellers (not shown) that are attached to shaft **24** and rotate when shaft **24** is rotated by energizing motor **18**. Rotating the impellers pressurizes fluid F drawn into pump **20**. Also in pump **20** are diffusers sequentially located by impellers, the combination of which direct the fluid F along a helical path inside pump **20**. Production tubing **28** is shown attached to an upper end of pump **20**, and pressured fluid is discharged from pump **20** into tubing **28**. Tubing **28** in shown supporting the ESP **10** within wellbore **12**. An upper end of tubing **28** attaches to a wellhead assembly **30** mounted on surface S. Wellhead assembly **30** provides pressure control for well **12** and way for diverting the produced liquid to offsite facilities for further processing or storage. In the example shown, a power cable **32** extends down to motor **18** within well **12** and which conducts electricity from a power source **34** on surface S to motor **18**. A communication means **36** is optionally included for conveying command signals and operational data between surface S and the ESP **10**; examples of communication means **36** include wireless, hardwired, a telemetry means, and combinations.
(11) FIG. **2** is a side sectional view of a portion of seal section **22** in ESP **10**. Provided in this example is that seal section **22** has an outer housing **38** that circumscribes an axis Ax of seal section **22**. Mounted to an uphole end of housing **38** is a seal head **40**, which is shown as a substantially solid and cylindrical member. An annular labyrinth chamber **42** is formed within housing **38** and a head chamber **44** is formed inside seal head **40**. As described in more detail below, the head chamber **44** is selectively in communication with pressure and fluid ambient to the ESP **10**. A passage **46** is formed through the seal head **40** and extends from a downhole end of the head chamber **44** into an uphole end of the labyrinth chamber **42**. An elongated breather tube **48** is shown having an uphole end **50** attached to where passage **46** intersects labyrinth chamber **42**. A downhole end **52** of breather tube **48** is in the labyrinth chamber **42** and so that chambers **42**, **44** are in communication with one another via passage and breather tube **48**. A mid portion of breather tube **48** is configured with a loop **54**, shown as a U-shaped portion, and that increases an effective length of the breather tube **48** past its actual axial length shown within seal section **22**.
(12) Inside housing **38** is an annular guide tube **56** that extends from the downhole portion of the seal head **40** and attaches to an uphole side of a bulkhead **58**, where bulkhead **58** is illustrated as a cylindrical member with an axial bore **59** and attached to a downhole end of housing **38**. Guide tube **56** circumscribes shaft **24** and defines an inner radial boundary of the labyrinth chamber **42**. A port **60** is formed radially through a side wall of guide tube **56** at a location axially uphole from the downhole end **52** of breather tube **48**. The guide tube **56** forms a guide tube chamber **62** within and that is in pressure communication with motor **18** (FIG. **1**). A bore **63** is also formed axially through seal head **40** and shaft **24** extends axially through bore **59** and bore **63**. Inner and outer bearings **64**,

66 are shown in between shaft **24** and bore **63**. A mechanical face seal **68** is illustrated within the head chamber **44** and around shaft **24**, and that defines a barrier to pressure and fluid communication through bore **63** along shaft **24**. Also formed through seal head **40** is a vent blind **70** that extends axially from a downhole end of the seal head **40** and then projects radially inward to bore **63** between bearing **64**, **66**, and the mechanical face seal **68**.

(13) Still referring to FIG. **2**, a pressure communication path P is shown that schematically depicts a route of pressure communication between ambient hydrostatic pressure and dielectric fluid inside motor **18** (FIG. **1**). As noted above, chamber **44** is in communication with ambient hydrostatic pressure, path P is shown extending from chamber **44** through passage **46** and breather tube **48** into the labyrinth chamber **42**, then across port **60** and into guide tube chamber **62**, which communicates directly with the dielectric fluid in the motor **18**. In the example shown, the spatial locations of the downhole end **52**, the length of the labyrinth chamber **42**, and the length between the downhole end **52** and port **60** form an elongated flow path for any well fluid that may enter into the head chamber **44** and motor **18**. Moreover, the addition of the U-shaped loop **54** inside breather tube **48** further lengthens this communication path to increase effectiveness of the barrier to communicating to any damaging well fluid to the motor **18**. In an alternative, breather tube **48** is configured to have a diameter that increases with distance away from the uphole end **50** which further introduces a resistance to flow of well fluid towards motor **18** and through the fluid communication path P.

(14) Referring out of FIGS. **2**A-**2**C, an alternative example of the breather tube **48**A is shown. In this example, the breather tube **48**A is equipped with a series of flow loops **72**A.sub.1-n are arranged along the breather tube **48**A and which an uphole and downhole ends **73**A.sub.1-n and **74**A.sub.1-n that intersect the breather tube **48**A at axially spaced are locations. In an example, the breather tube **48**A with the flow loops **72**A.sub.1-n define a valvular conduit, an example of which is shown in Tesla, U.S. Pat. No. 1,329,559, which is incorporated herein in its entirety and for all purposes. In the example FIGS. **2**A-**2**C, the flow loops **72**A.sub.1-n are disposed generally sequential with one another along the length of the breather tube **48**A, alternatives exist where there is some overlapping between these flow loops **72**A.sub.1-n or there is some axial distance between a downhole end **72**A.sub.1-n and the next sequential uphole end **73**A.sub.1-n of the adjacent flow loops **72**A.sub.1-n.

(15) Referring specifically to FIG. **2**B, an example of a flow of fluid F F.sub.48A inside breather tube **48**A is shown in a direction that is downhole, i.e., from the uphole end **50**A to downhole end **52**A. As shown, a main flow of fluid F.sub.48A flows inside the breather tube **48**A which entered tube **48**A from the uphole end **50**A. Portions of the main flow of fluid F.sub.48A selectively divert into bypass **72**A.sub.1, shown as flow of fluid F.sub.72A1, and due to the high angle of bending within the flow loop **72**A.sub.1, flow of fluid F.sub.72A1 reenters tube **48**A at a location downstream of where it was diverted from tube **48**A and in a direction that is substantially opposite that of flow of fluid F.sub.48A The cumulative effect of the multiple flow loops **72**A.sub.1-n retards the flow of fluid F.sub.48A in this direction of from uphole end **50**A to downhole end **52**A. This is further schematically illustrated by the relative orientations of axis A.sub.48A with axes A.sub.73A, A.sub.74A. As shown, angles Φ.sub.1, Φ.sub.2 between axis A.sub.48A and axes A.sub.73A, A.sub.74A are each acute angles, and in examples these angles have similar or the same magnitudes. The schematic example illustrated how the bypass loops **72**A.sub.1-n draw off a side stream of fluid at an angle that is then redirected back into the main flow at substantially the same angle and in a direction that is largely opposite. Further shown in FIG. **2**B is the flow path FP which illustrates the direction of flow of fluid F.sub.48A from uphole end **50**A to downhole end **52**A.

(16) FIG. **2**C illustrates an example of a flow of fluid F.sub.48A through tube **48**A in a direction from downhole end **52**A to uphole end **50**A. In this example, fluid such as dielectric fluid in motor **18** is flowing uphole and towards the seal head **40** and is in contrast to the flow in the opposite

direction which could bring damaging well fluid to the motor **18**. In this example, angles ϕ.sub.1, ϕ.sub.2 between axis A.sub.48A and the axes of uphole and downhole ends A.sub.73A, A.sub.74A are both obtuse. In examples, the magnitudes of angles ϕ.sub.1, ϕ.sub.2 are similar or equal. In the example illustrated in FIG. **2**C, side stream magnitudes are negligible as the geometry of the breather tube **48**A discourages fluid diversion from the main flow of fluid F.sub.48A into the bypass loops **72**A.sub.1-n. Because the obtuse angles required for the fluid to maneuver before entering the flow loops **48**A creates or allows for a very small amount of side stream fluid, so that flow in the direction from the downhole end **52**A to uphole end **50**A is largely unimpeded along breather tube **48**A.

(17) FIG. **3** is an alternate embodiment of the portion of seal section **22**B making up the ESP **10**B. In this example, port **60**B is formed radially through a downhole portion of guide tube **56**B and inside housing **38**B is an annular labyrinth wall **75**B is provided that has a downhole end fixed to the downhole portion of guide tube **56**B and spaced radially outward from port **60**B. Labyrinth wall **75**B is spaced radially outward from a mid-portion of the guide tube **56**B and has an upper end that is adjacent to a shroud **76**B. Shroud **76**B projects radially outward from guide tube **56**B and has a lip **77**B on its radial tip that extends downhole past the upper end of labyrinth wall **75**B. In this example, a flight member **78**B is included which is a sleeve-like element that is shown circumscribing shaft **24** and mounted onto an outer surface of shaft **24**B, such as with a key (not shown) for rotational coupling, and in alternatives is axially coupled with circlips or by sleeves (not shown) that extend to bearings **64**B, **66**B. Flight member **78**B is shown in FIG. **3** extending along an axial distance largely circumscribed by the labyrinth wall **75**B. Included on an outer surface of flight member **78**B is a flight element **80**B that projects radially outward from flight member **78**B and extends along flight member **78**B in a helical pattern. An optional outer flight **82**B is included which is formed along an inner surface of guide tube **56**B and that faces an outer surface of flight member **78**B. On an inner surface of outer flight **82**B is an outer flight member **84**B, which extends axially along a portion of outer flight **82**B and that projects radially inward towards flight element **80**B; and in examples is configured complementary to flight element **80**B. Webs **86**B are shown in FIG. **3**A extending radially between diffuser flight **82**B and labyrinth wall **75**B and through a flow path **88**B formed between flight **82**B and wall **75**B. In a non-limiting example guide tube **56**B (with its outer flight **82**B and outer flight member **84**) and labyrinth wall **75**B are a single piece and held together with the webs **86**A. In examples, the single piece is cast or fabricated using additive manufacturing.

(18) In a non-limiting example of operation, flight member **78**B and flight element **80**B rotate with rotation of shaft **24**B and generate a flow of fluid along shaft **24**B in an uphole direction within the guide tube **56**B. Past the bearings **64**B, **66**B the fluid impinges on a downhole surface of the mechanical seal **68**B and generates a localized increase in pressure, which in the event of failure of mechanical seal **68**B would form a barrier to a flow of well fluid along shaft **24**B and to motor **18** (FIG. **1**).

(19) The present invention described herein, therefore, is well adapted to carry out the objects and attain the ends and advantages mentioned, as well as others inherent therein. While a presently preferred embodiment of the invention has been given for purposes of disclosure, numerous changes exist in the details of procedures for accomplishing the desired results. These and other similar modifications will readily suggest themselves to those skilled in the art, and are intended to be encompassed within the spirit of the present invention disclosed herein and the scope of the appended claims.

Claims

1. An electrical submersible pumping system (“ESP”) that is selectively disposed in a wellbore, the ESP comprising: a motor; a shaft having an end attached to the motor; a pump attached to an end of

the shaft distal from the motor; and a seal section disposed between the motor and the pump comprising, a breather tube having, an uphole end that is in pressure communication to ambient hydrostatic pressure, a downhole end distal from the uphole end and that is in pressure communication with the motor, and a mid-portion having an undulating contour that comprises a recycle loop mounted to the breather tube, the recycle loop having a curved mid-section and upper and lower ends that intersect the breather tube at spaced apart locations along a length of the breather tube to define a valvular conduit.
2. The ESP of claim 1, wherein the undulating contour comprises a U-shaped bend.
3. The ESP of claim 1, wherein the upper end intersects the breather tube at an upper intersection and the lower end intersects the breather tube at a lower intersection that is upper intersection, wherein the portion of the recycle loop proximate to the upper intersection is at an acute angle to a portion of the breather tube adjacent the upper intersection and on a side opposite the uphole end, and wherein the portion of the recycle loop proximate to the lower intersection is at an acute angle to a portion of the breather tube adjacent the lower intersection and on a side opposite the uphole end.
4. The ESP of claim 3, further comprising a multiplicity of recycle loops arranged in series along the length of the breather tube.
5. The ESP of claim 1, further comprising a seal section housing, a guide tube in the seal section that circumscribes a portion of the shaft and that is spaced radially inward from the seal section housing, and a labyrinth chamber defined in an annulus between the guide tube and seal section housing.
6. The ESP of claim 5, wherein the breather tube is disposed in the labyrinth chamber.
7. The ESP of claim 6, wherein a port is formed radially through the guide tube at a location axially between the uphole and downhole ends of the breather tube, so that a pressure communication path between the pump and the motor extends from the uphole end of the breather tube to the downhole end of the breather tube, uphole to the port, and downhole to the motor in an annular space between the shaft and the guide tube.
8. The ESP of claim 1, further comprising a mechanical seal on the shaft and a helical flight on-the-downhole of the mechanical seal, the helical flight configured to increase fluid pressure on a downhole side of the mechanical seal.
9. The ESP of claim 1, wherein the undulating contour comprises a diameter of the breather tube that increases with distance from the uphole end to the downhole end.
10. An electrical submersible pumping system ("ESP") selectively disposed in a wellbore, the ESP comprising: a motor; a pump uphole of the motor; a shaft having an end attached the motor and an opposing end attached to the pump; and a seal section disposed between the motor and the pump having a labyrinth chamber with a flow path that restricts fluid communication, the flow path extending through a breather tube with a diameter that creases with distance downhole.
11. The ESP of claim 10, further comprising a helical flight on an outer surface of the shaft that operates as a positive displacement pump with rotation of the shaft and so that fluid exiting the positive displacement pump generates an increase in pressure on a downhole side of a mechanical seal around the shaft.
12. The ESP of claim 11, wherein the seal section comprises an outer housing with a seal head on an uphole end of the housing, the seal head having an axial bore that receives the shaft, and wherein a vent tube is formed through the seal head that provides a flow path for fluid to flow into the seal section from adjacent the mechanical seal.
13. The ESP of claim 10, wherein the flow path extends through a breather tube with a U-shaped portion.
14. The ESP of claim 10, wherein the fluid flow path extends through a valvular conduit.
15. The ESP of claim 14, wherein fluid flow in a direction to the motor is restricted in the valvular conduit, and wherein fluid flow in a direction away from the motor flows substantially unimpeded

through the valvular conduit.

16. A method of operating an electrical submersible pumping system ("ESP") comprising: providing electricity to the ESP is disposed in a wellbore, to energize a motor in the ESP to rotate a shaft connected between the motor and a pump, so that fluid in the wellbore is pressurized by the pump; equalizing pressure of dielectric fluid inside the motor with hydrostatic pressure in the wellbore by communicating pressure of the wellbore to the dielectric fluid through a communicating fluid that is flowable along a pathway that intersects a seal disposed between the motor and pump; and blocking fluid in the wellbore from communicating to dielectric fluid inside the motor by diverting a portion of the communicating fluid flowing towards the motor and reintroducing the portion into the pathway in a direction pointing away from the motor.

17. The method of claim 16, wherein the communicating fluid is diverted from the pathway at a first location and reintroduced into the pathway at a second location, wherein the second location is between the first location and the motor.

18. The method of claim 16, wherein the diverted fluid flows along a flow loop, the method further comprising communicating dielectric fluid from the motor along the pathway through the seal, diverting a portion of the dielectric fluid through the flow loop, and reintroducing the diverted dielectric fluid back into the pathway in a direction pointed away from the motor.